1 2D-DCT Transform

(a)

For $A^{4\times4}$, N=4. Therefore, the DCT matrix T should be:

$$T[i,j] = \begin{cases} \sqrt{\frac{1}{N}} = \frac{1}{2}, & \text{if } i = 0 \\ \sqrt{\frac{2}{N}} \cos\left(\frac{(2j+1)i\pi}{2N}\right) = \frac{1}{\sqrt{2}} \cos\left(\frac{(2j+1)i\pi}{8}\right), & \text{if } 0 < i < N,\ 0 \le j < N \end{cases}$$

.

$$T = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2}\cos\frac{\pi}{8} & \frac{1}{2}\cos\frac{3\pi}{8} & \frac{1}{2}\cos\frac{5\pi}{8} & \frac{1}{2}\cos\frac{7\pi}{8} \\ \frac{1}{2}\cos\frac{2\pi}{8} & \frac{1}{2}\cos\frac{6\pi}{8} & \frac{1}{2}\cos\frac{10\pi}{8} & \frac{1}{2}\cos\frac{14\pi}{8} \\ \frac{1}{2}\cos\frac{3\pi}{8} & \frac{1}{2}\cos\frac{9\pi}{8} & \frac{1}{2}\cos\frac{15\pi}{8} & \frac{1}{2}\cos\frac{21\pi}{8} \end{bmatrix} \approx \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.653281 & 0.270598 & -0.270598 & -0.653281 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.270598 & -0.653281 & 0.653281 & -0.270598 \end{bmatrix}.$$

As for the DCT transform consequence $D = TAT^T$

Therefore, the consequence D= $\begin{bmatrix} 10.0000 & 9.2388 & 0.0000 & -3.8268 \\ 9.2388 & 8.5355 & 0.0000 & -3.5355 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ -3.8268 & -3.5355 & 0.0000 & 1.4645 \end{bmatrix}$.

(b)

Comparing with the 2D-DFT, the 2D-DCT is easier to compute. The DCT only consider the cosine computation. The DFT must consider cosine and sine computation at same time. They will have similar performance on sjowing the color frequency on images. However, the DCT could save half of computation cost. Therefore, DCT performs better from my perspective.

2 Controllable Vedio Generation

Part 1:

- **Method Description:** Cross-subject motion-guided video generation aims to transfer human motion dynamics into realistic animal behaviors through pose-guided diffusion models.
- **Objective:** The objective is to generate biologically consistent videos where animals perform human-like actions naturally while preserving each species' motion traits.
- **Motivation:** This method would reduce the production cost of animal films and animations, enable controllable motion synthesis across species, and enhance visual realism and creativity in AI-assisted video generation. *(95 Words)*

Part 2:

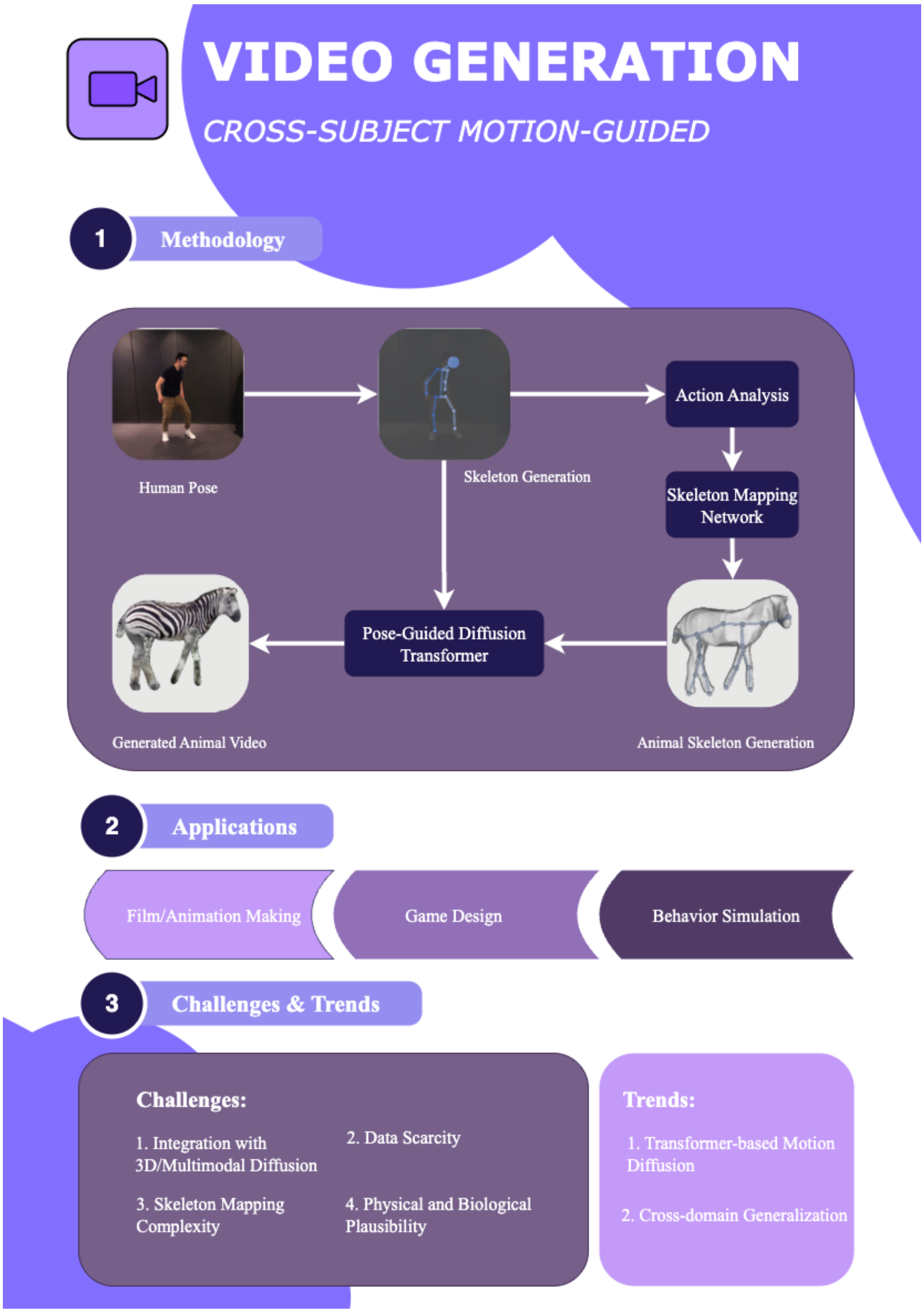

Figure 1: Infographic of Cross-Species Motion Transfer

[1] [2] [3]

References

[1] Q. Gan, Y. Ren, C. Zhang, Z. Ye, P. Xie, X. Yin, Z. Yuan, B. Peng, and J. Zhu, "HumanDiT: Pose-Guided Diffusion Transformer for Long-form Human Motion Video Generation," Aug. 2025. arXiv:2502.04847 [cs].

[2] K. Sun, D. Litvak, Y. Zhang, H. Li, J. Wu, and S. Wu, "Ponymation: Learning Articulated 3D Animal Motions from Unlabeled Online Videos," July 2024. arXiv:2312.13604 [cs].

[3] Z. Yang, W. Zhu, W. Wu, C. Qian, Q. Zhou, B. Zhou, and C. C. Loy, "TransMoMo: Invariance-Driven Unsupervised Video Motion Retargeting," Apr. 2020. arXiv:2003.14401 [cs].